

COURSE:
SUBJECT:
PAPER CODE:

BCA
Discrete Mathematics
BCA101

SEMESTER: I
M.MARKS: 30
TIME: 1 Hour 30 Minutes

NOTE: - Part - A is compulsory. Attempt any 2 questions from part B

PART - A

Q. No.	CO	Question	Marks
1.	1	(i) Draw the Hasse Diagram of (a) The Poset $(S,)$ where $S = \{1, 2, 3, 4, 6, 8, 9, 12, 18, 24\}$ (b) The Poset $(P(S), \subseteq)$ where $S = \{1, 2, 3\}$	2.5
	1	(ii) Define the symmetric difference of two sets A and B. If $A = \{1, 2, 3, 4, 5\}$; $B = \{3, 4, 7, 8, 9\}$; $C = \{6, 10, 9, 11\}$ Find $A \Delta B$ and $B \Delta C$.	2.5
	2	iii. Find the domain and range of the following functions (a) $f(x) = (x) / (x+1)$ (b) $f(x) = \sin 2(x^2) + \cos 2(x^2)$	2.5
	2	iv. Let R denote the set of all real numbers and $f: R \rightarrow R$ be defined as $f(x) = 3x + 9 \forall x \in R$. Show that f is both one-to-one and onto function. Find the formula for $f^{-1}: R \rightarrow R$.	2.5

PART - B

Q. No.	CO	Question	Marks
2	1	(i) In a group of 60 students, 25 play cricket, 30 play football, 24 play hockey, 10 play cricket and football, 9 play cricket and hockey, 12 play hockey and football and five play all the three. Find out: (a) How many play only one game (b) How many play no game. (ii) Prove that the relation of "Divisibility" is a Poset on N (set of natural numbers). Is the relation of divisibility a poset on the set of Integers Z? Justify your answer	10
3	2	i. Draw the Hasse Diagram for D_{60} under divisibility relation and find the following: (a) Maximal, Minimal, Greatest, Least elements (b) Atoms, Join Irreducible elements (c) Show that the lattice D_{60} is distributive (d) Supremum of $A = \{2, 3, 12\}$ (e) Infimum of $B = \{2, 4, 15\}$ ii. Let $A = \{1, 2, 3, 4, 5, 6\}$ Let R be a relation defined on A as $R = \{(x, y) : x+y \text{ is a divisor of } 24\}$ (a) Find the matrix of the relation M_R (b) Find the Domain and Range of R (c) Find the inverse relation R^{-1} (d) Directed graph of R	10
4	2	i. Write the converse, inverse and contrapositive statement of the following statement. "India wins whenever Virat Kohli scores Century". ii. State De-Morgan Laws in Sets. Prove any one. Verify the same using truth table	10

VIVEKANANDA INSTITUTE OF PROFESSIONAL STUDIES- TC
CLASS TEST - NOV -DEC - 2022

COURSE: BCA
SUBJECT: Technical Communication
PAPER CODE: BCA 109

SEMESTER: I
M.MARKS: 30
TIME: 1 Hour 30 Minutes

NOTE: - Part - A is compulsory. Attempt any 2 questions from part B

PART - A

10 Marks

Attempt any two of the given questions

Q. No.	CO	Question	Marks
1.		Write short notes on:	2*5
	CO1, CO3	a) What is technical communication & explain its features in detail.	5
	CO2, CO4	b) Explain the merits & demerits of oral communication.	5
	CO1, CO3	c) Explain the communication process and its elements in detail.	5

PART - B

2*10 = 20 Marks

Q. No.	CO	Question	Marks
2	CO1, CO3	What are the seven golden principles of communication?	10
3	CO2, CO4	Explain the types and styles of interviews in detail.	10
4		Write short notes on:	2*5
	CO2, CO4	a) Styles of oral communication	5
	CO1, CO3	b) Computer-aided technical communication	5

VIVEKANANDA INSTITUTE OF PROFESSIONAL STUDIES- TC
CLASS TEST – NOV -DEC - 2022

COURSE:
 SUBJECT:
 PAPER CODE:

BCA
 WEB TECHNOLOGIES
 BCA 107

SEMESTER: FIRST
 M.MARKS: 30
 TIME: 1 Hour 30 Minutes

NOTE: - Part – A is compulsory. Attempt any 2 questions from part B

PART – A			10 Marks
Q. No.	CO	Question	Marks
1.	CO1	Differentiate between Client-side and server-side scripting Language	2.5
2.	CO1, CO3	Difference between container tag and empty tag with 3 tags each.	2.5
3.	CO3	How to implement HTML Table? Differentiate between rowspan and colspan in a table.	2.5
4.	CO1, CO3	Different types of HTML list with example.	2.5

PART – B		2*10 = 20 Marks							
Q. No.	CO	Question	Marks						
2	CO1, CO3	Write the HTML code to generate a Web Page. Consider the following constraints while writing the HTML code: i) Title of the page should be "Hotel Industry". Background colour of the page should be "Yellow", Link colour should be "Blue", visited link colour should be "Red" ii) Image used in the page is the file "hill.jpg" with height and width 250 by 250 pixel. iii) Table below should have a caption "ABOUT" and border of width 2. <table><tr><th>Resorts</th><th>Hotels</th></tr><tr><td>Beach Resorts</td><td>Business Hotels</td></tr><tr><td>Hill Resorts</td><td>Leisure Hotels</td></tr></table> iv) Webpages are hyper-linked to the list items in the given list: • Front Office as "front.html" • Food & Beverages as "food.html"	Resorts	Hotels	Beach Resorts	Business Hotels	Hill Resorts	Leisure Hotels	10
Resorts	Hotels								
Beach Resorts	Business Hotels								
Hill Resorts	Leisure Hotels								
3	CO1, CO3	a) Discuss frame in HTML? What are the benefit of using frames in a webpage? Also explain the floating frame with example. Write HTML Code to generate the following frame and link each frame with different web pages: <table><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>							
4	CO1, CO3	a) What are Cascading Style Sheets (CSS)? Explain the different types of style sheets with the help of HTML code. b) Design an HTML Form for student that include the following elements: Student's name, Qualification, Gender, Mobilenos, Address, Hobbies. Also make buttons to submit and clear the data.							

COURSE: BCA

SUBJECT: Fundamentals of Computers and IT

PAPER CODE: BCA105

SEMESTER: I

M.MARKS: 30

TIME: 1 Hour 30 Minutes

NOTE: - Part - A is compulsory. Attempt any 2 questions from Part B**PART - A**

10 Marks

Q. No.	CO	Question	Marks
1.	CO1 CO2	Answer any 4 of the following: (a) Explain the important characteristics of a Computer system. (b) What is the difference between data and information? Explain the term data processing. (c) Describe applications of a Computer system. (d) What is meant by the base of the number system? Distinguish between binary & hexadecimal number systems. (e) Differentiate RISC and CISC processors.	(2.5 * 4 = 10)

2 * 10 = 20 Marks

PART - B

Q. No.	CO	Question	Marks
2	CO1	(a) Explain the basic organization of a Computer system with the help of a block diagram. (b) Explain the primary or main memory of a Computer system. Differentiate between RAM & ROM?	(5 * 2 = 10)
3	CO2	(a) Convert the following: (i) 423 from base 10 to base 2. 11010011 (ii) 1101010 from base 2 to base 16. 6A (iii) A16 from base 16 to base 10. 2582 (iv) 4675 from base 10 to base 8. (11103)8 (v) 127 from base 8 to base 2. 1010111	(5 * 2 = 10)
4	CO1	(a) Explain various generations of the Computer system with their key technologies and characteristics. (b) What are the registers? Explain MAR, MBR, PC & IR.	(5 * 2 = 10)

VIVEKANANDA INSTITUTE OF PROFESSIONAL STUDIES- TC
CLASS TEST - NOV -DEC - 2022

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COURSE: BCA
SUBJECT: Programming Using 'C' Language
PAPER CODE: BCA 103

SEMESTER: I
M.MARKS: 30
TIME: 1 Hour 30 Minutes

NOTE: - Part - A is compulsory. Attempt any 2 questions from part B

PART - A

10 Marks

Q. No.	CO	Question	Marks
1.	1	Explain the difference between while and do-while with the help of example	2
2.	1	Discuss the hierarchy of the various types of operators in C.	2
3.	2	Explain the use of break, goto and continue statements.	2
4.	1	Write a program to print sum of squares of first n natural numbers.	2
5.	1	Write the output of following code: <pre>#include <stdio.h> int main() { int a = 1, b = 1, c; c = a++ + b; b = ++c - b; printf("%d, %d, %d ", a, b, c); }</pre>	2

PART - B

2*10 = 20 Marks

Q. No.	CO	Question	Marks
2	1,2	A. Explain the difference in call by value and call by reference using an example.	5+5
3	1	B. Explain bit-wise operators with example. A. Discuss the conditional statements in C with all variations, using examples. B. Write a program to generate the given pattern using for loop <pre>* * * * * * * * * * </pre>	5+5
4	1	A. What are storage classes of variables? Explain with example. B. Write a program to check if the given number is prime or not.	5+5